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SCHOOL OF COMPUTING

Department of Computer Science and Engineering with specialization in Data Science

Interdisciplinary Project

Review 1

Ai- powered resume based interview simulator

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1. Abstract

project presents an intelligent AI-based interview simulation system that analyzes a candidate's resume and generates personalized interview questions. The system leverages Natural Language Processing (NLP) and Large Language Models (LLMs) to extract key information such as skills, experience, and projects from the resume. Based on this analysis, the system dynamically creates relevant technical and behavioral interview questions tailored to the candidate's profile.

The platform aims to provide a realistic interview experience by adapting questions in real-time and evaluating user responses. This helps candidates identify their strengths and weaknesses, improve their communication skills, and prepare effectively for real-world interviews. The system can be extended to include feedback generation, scoring mechanisms, and domain-specific interview simulations.

2. Introduction

Candidates lack personalized interview preparation. Existing platforms provide generic questions that do not align with individual resumes.

This project addresses this gap by introducing an AI-driven interview system that personalizes the interview experience based on the candidate's resume. By analyzing the resume content, the system identifies relevant topics and generates context-aware questions. Unlike generic interview platforms, this system ensures that the questions are directly aligned with the candidate's skills, projects, and experiences.

3. Problem Statement

Many job candidates struggle with interview preparation due to a lack of personalized guidance and realistic practice environments. Existing platforms often provide generic questions that do not reflect the candidate's actual profile, leading to ineffective preparation.

There is a need for an intelligent system that:

- Understands and analyzes a candidate's resume
- Generates customized interview questions based on individual profiles
- Simulates a real interview experience
- Provides meaningful feedback to improve performance

The challenge lies in accurately extracting relevant information from resumes and generating contextually appropriate and diverse interview questions using AI techniques.

4. Objectives

- **resumes using NLP techniques**
- To generate **personalized interview questions** based on extracted information
- To simulate a **real-time interview experience** using AI
- To evaluate user responses and provide **constructive feedback**
- To support multiple domains (e.g., Data Science, Web Development, etc.)
- To improve user confidence and communication skills
- To design a scalable system that can be integrated into career platforms

6. Proposed System

- **System Overview**
  The Academic Knowledge Chatbot is an AI-based system designed to assist students by answering academic queries. It uses Generative AI to understand user questions and provide accurate, context-aware responses by retrieving information from academic resources like notes and PDFs.

- **Technologies used**

This project uses Generative AI technologies along with tools for data processing, integration, and user interface design to build an intelligent academic chatbot.

7. Tools and Technologies

Category	Tools
Programming	Python
Framework	FastAPI
Embeddings	OpenAI embaddings/Sentence transformers
Vector Database	FAISS
LLM(AI Model)	OpenAi/LLaMA
Data Source	User resume
Platform	Web app/ React
Libraries	PyMuPDF, langchain, numpy, pandas

8. References

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CONCLUSION

- The AI Resume-Based Interview System successfully demonstrates how Artificial Intelligence can be used to personalize interview preparation. By analyzing resumes and generating context-aware questions, the system provides a more realistic and effective practice environment compared to traditional methods.
- This project highlights the practical use of NLP, embeddings, and LLMs in solving real-world problems. It helps users improve their technical knowledge, communication skills, and confidence.
- In the future, the system can be enhanced with features such as voice-based interaction, real-time feedback scoring, and adaptive questioning to further simulate real interview scenarios.